MODULE 11: Mini-Project Design and Documentation

Secure-the-Cell Case Study

MINI-PROJECT OVERVIEW

What This Module Is About

In This Module, You Will:

- Integrate ideas from all previous modules into a cohesive project
- Work on the Secure-the-Cell mini-project using real-world principles
- Focus on clarity, safety, and professional documentation standards
- Prepare a professional one-slide presentation that tells your complete story

Core Focus Areas

Primary Focus: Critical thinking, analytical reasoning, and clear communication of security concepts

Not the Focus: Attack simulations, exploit development, or complex penetration testing tools

This module emphasizes defensive understanding and responsible analysis over offensive techniques.

LEARNING OBJECTIVES

Purpose of the Mini-Project

This mini-project bridges theory and practice, allowing you to demonstrate your understanding of OT security principles in a realistic scenario. You'll develop skills that translate directly to professional security analysis work.

Apply OT Security Concepts

Use knowledge from previous modules to analyze a small but realistic industrial control system

Practice Asset Identification

Build comprehensive inventories and develop risk-based thinking for critical assets

Interpret Monitoring Data

Analyze system behavior responsibly, drawing evidence-based conclusions from real monitoring data

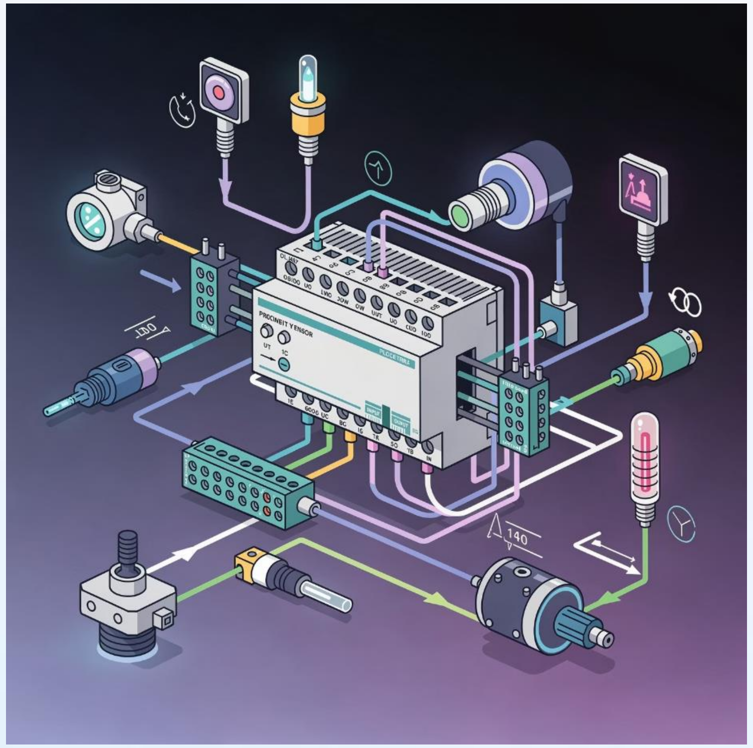
Explain Findings Clearly

Communicate complex security observations in limited time with professional clarity

Expected Outcome: Real-world style reasoning and professional documentation skills, not just theoretical answers. You'll learn to think like a security analyst working in operational technology environments.

What Is "Secure-the-Cell"?

The System You Will Analyze



The Secure-the-Cell case represents a small industrial control cell that mirrors real manufacturing or process control environments. This controlled learning scenario provides a safe space to apply security analysis techniques.

System Components

- **Field devices:** Sensors measuring physical conditions and actuators controlling operations
- **One controller:** PLC or embedded controller managing automation logic
- **One operator interface:** HMI (Human-Machine Interface) for monitoring and control
- **Monitoring data:** Provided datasets in CSV or log format showing system activity

1

Realistic Scale

Small enough to analyze thoroughly, large enough to demonstrate real concepts

2

Safe Environment

Controlled case study with no risk to actual operational systems

3

Complete Dataset

All necessary information provided for comprehensive analysis

SCOPE DEFINITION

Scope and Boundaries

What Is Included and Excluded

✓ Included in Your Analysis

Asset Details

Complete information about devices, their roles, and criticality in the system

Network Behavior Data

Communication patterns, timing, and frequency of device interactions

Monitoring Observations

System logs and event data showing normal and potentially anomalous activity

✗ Not Included

Live Systems

No connection to actual operational environments or active industrial networks

Exploit Techniques

No vulnerability exploitation, attack tools, or offensive security methods

Attack Simulations

No penetration testing or active security testing against systems

Project Goal: Defensive understanding and responsible analysis, not exploitation or aggressive testing

Understanding the Dataset

What the Data Represents

The monitoring dataset provides a window into system behavior over time. Learning to read and interpret this data accurately is a fundamental skill for OT security professionals.

Timing Information

When communication events occurred, including timestamps and frequency patterns

Device Communication

Which devices communicated with each other and the nature of their interactions

Message Counts

Volume of communications, helping identify normal patterns and anomalies

Protocol Types

Basic protocol or event type information showing communication methods

Your Analysis Approach

- Look for communication patterns that define normal behavior
- Identify deviations from expected patterns
- Consider operational context for observations
- Document findings with evidence

Analysis Boundaries

You will focus on high-level behavior analysis and pattern recognition. This project **avoids** deep packet-level inspection or protocol reverse engineering.

The goal is operational security awareness, not network forensics.

TASK 1

Organize Project Files

What You Will Prepare

Professional project organization demonstrates attention to detail and makes your work reviewable. A clear structure helps instructors, peers, and future employers understand your analytical process.

1. **/data – Original Dataset**
   Store the original dataset here as read-only. Never modify source data. Keep pristine copies for verification and reference.

2. **/analysis – Notes and Summaries**
   Working documents, preliminary observations, calculations, and draft analysis. Your thinking process lives here.

3. **/inventory – Asset and Risk Sheets**
   Formal asset inventory spreadsheets and risk register documents. These become reference materials.

4. **/findings – Written Observations**
   Finalized analysis write-ups, monitoring summaries, and documented conclusions with supporting evidence.

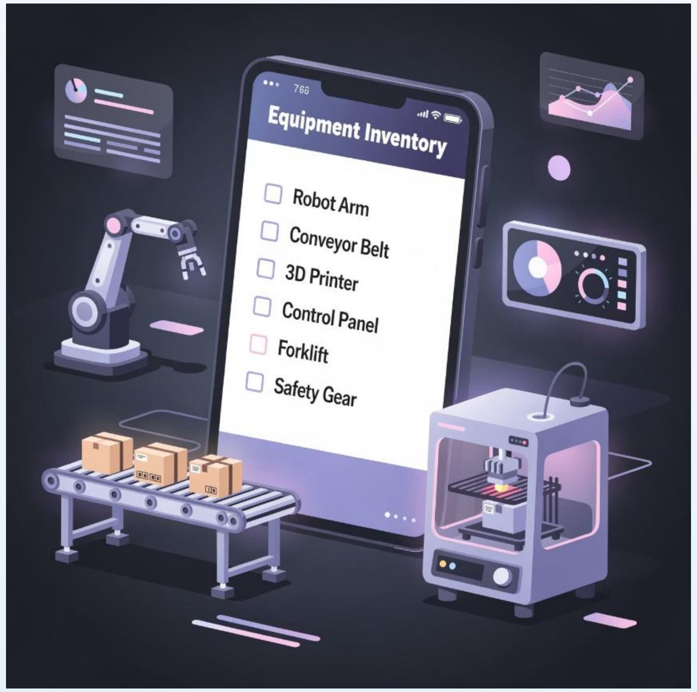
5. **/slide – Final One-Slide Summary**
   Your completed presentation slide and any supporting graphics or charts used in the final deliverable.

Expected Outcome: A professional, review-ready project structure that any stakeholder can navigate and understand. Clean organization reflects clear thinking.

TASK 2

Build the Asset Inventory

What You Will Identify



Asset inventory forms the foundation of security analysis. You cannot protect what you don't know exists. Each asset must be documented with enough detail for risk assessment.

For Each Asset, Record:

- **Asset Name and Type**

  Clear identifier and classification (e.g., "Temperature Sensor TS-01" or "Main PLC Controller")

- **Role in the System**

  What function this asset performs and why it exists in the control cell

- **Criticality Level**

  Rate as High, Medium, or Low based on impact to operations if compromised or unavailable

- **Ownership**

  Assumed responsible party (e.g., Operations Team, Engineering, IT/OT Security)

Expected Outcome: Clear visibility of what exists in the system, enabling informed risk decisions and security planning. Your inventory should answer the question: "What are we protecting?"

TASK 3

Create the Risk Register

How You Will Think About Risk

Risk assessment in OT environments requires balancing technical vulnerabilities with operational realities. Your risk register documents potential security concerns and establishes priorities for mitigation.

1 **Identify Risk**

For each key asset, note one realistic risk based on its role and exposure

2 **Assess Likelihood**

Rate probability as Low, Medium, or High using reasonable assumptions

3 **Evaluate Impact**

Rate consequences as Low, Medium, or High considering safety and operations

4 **Propose Mitigation**

Suggest one reasonable control or countermeasure to reduce the risk

Example Risk Entry

Asset	Main PLC Controller
Risk	Unauthorized configuration change
Likelihood	Medium
Impact	High
Mitigation	Implement role-based access control

Focus on Justification

Your risk ratings don't need to be perfect—they need to be *justified*. Explain your reasoning clearly.

Avoid arbitrary scoring. Connect each risk to asset criticality, threat exposure, and potential operational impact.

TASK 4

Analyze Monitoring Data

What Questions You Must Answer

Effective monitoring analysis answers fundamental questions using evidence from the dataset. Your analysis should be factual, supported, and focused on observable behavior rather than speculation.

What Happened?

Describe the observable event, pattern, or behavior in neutral, factual language. Stick to what the data actually shows.

When Did It Happen?

Provide specific timing: date, time, duration, or frequency. Temporal context helps establish patterns and anomalies.

Where Did It Occur?

Identify which device, network segment, or system component exhibited the behavior. Location matters in distributed systems.

Why Does It Matter?

Explain the operational or security significance. Connect observations to business impact, safety, or system integrity.

✗ Avoid

- Guessing attacker intent
- Blaming specific individuals
- Drawing unsupported conclusions
- Sensationalizing findings

✓ Do Include

- Evidence-based observations
- Neutral, professional language
- Logical reasoning
- Operational context

Remember

You're analyzing a learning case. Focus on demonstrating analytical thinking, not proving technical superiority.

TASK 5

Summarize Findings

What Your Findings Should Show

A strong findings summary communicates your analysis clearly and concisely. It demonstrates your ability to move from raw data to actionable insights while maintaining professional objectivity.

01

Normal Behavior Pattern

Establish what "normal" looks like for this system: regular communication intervals, expected device interactions, typical message volumes

02

Deviation or Observation

Identify one noteworthy deviation from the baseline: unusual timing, unexpected communication, or irregular patterns

03

Operational Explanation

Provide a possible operational reason for the observation: scheduled maintenance, configuration change, or normal operational variance

04

Relevance to Risk

Connect the observation to your risk register: Does it elevate concern? Does it suggest a mitigation worked? How does it inform security posture?

Language Guidelines: Use short, factual, neutral language only. Avoid emotional words, absolutes, or dramatic claims. Your goal is credible analysis, not attention-grabbing headlines.

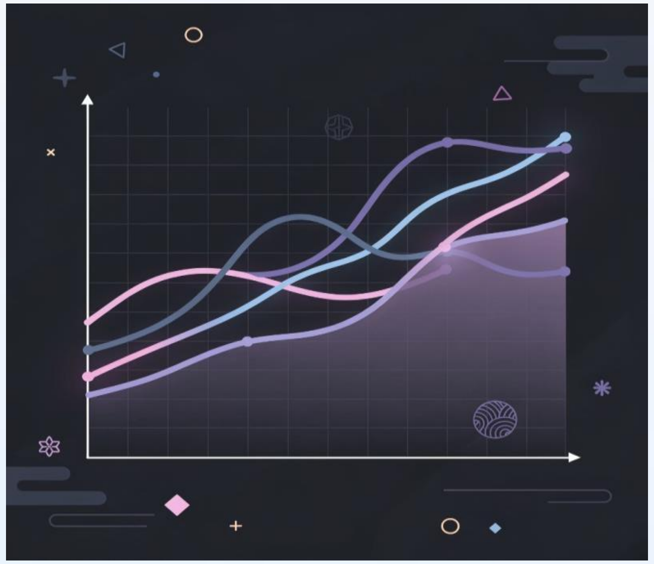
> "The HMI showed 15 connection attempts to the PLC between 02:00-02:15 on March 3, versus the normal 2-3 attempts per hour. This could indicate legitimate troubleshooting or automated backup, but warrants investigation given the PLC's high criticality."

TASK 6

Create Simple Visuals

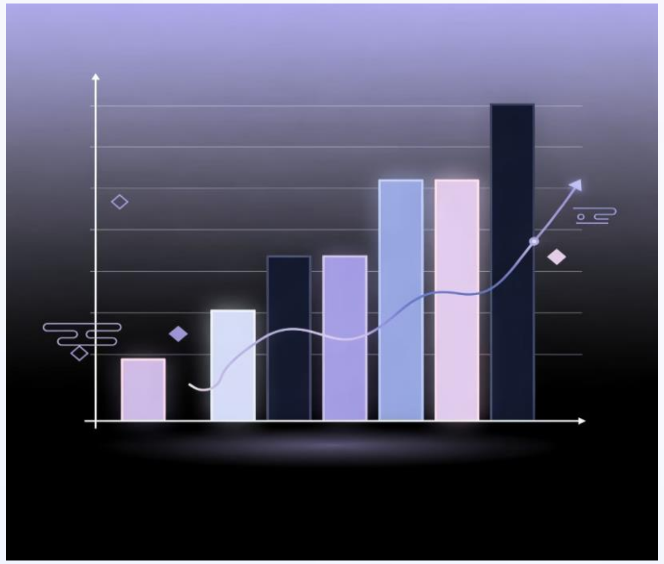
How You Will Show Data

Effective visuals support your narrative without overwhelming it. Choose the right chart type for your message, keep designs clean, and ensure every visual serves a clear purpose.



Line Chart → Trends Over Time

Use line charts to show how values change across time periods. Ideal for displaying communication frequency, message counts, or connection patterns throughout the day or week.



Bar Chart → Device Comparison

Use bar charts to compare different categories or devices. Perfect for showing relative activity levels, message volumes across devices, or comparing normal vs. observed values.

Table → Exact Values

Use tables when precise numbers matter or when showing multiple attributes for several items. Tables excel at presenting asset inventories, risk registers, or detailed findings summaries.

Design Rule: Visuals support explanation, not decoration. Every chart or table should answer a specific question or clarify a specific point. If you can't explain why a visual is there, remove it.

TASK 7

Design the One-Slide Story

Your Entire Project on One Slide

Your final presentation slide is the culmination of all your work. It must tell a complete, coherent story that can be understood in under two minutes. This is your opportunity to demonstrate both technical understanding and communication skill.

1. **System Context**

   Briefly describe what control cell you analyzed—what it does, key components, and its operational purpose

2. **Approach**

   Summarize your methodology: asset inventory → monitoring analysis → risk assessment. Show you followed a structured process.

3. **Key Findings**

   Present 2-3 most significant findings from your analysis. Include evidence and maintain neutral, professional language.

4. **Recommendations**

   Offer 1-2 practical, implementable recommendations based on your findings. Make them specific and actionable.

Design Principles

- Use clear visual hierarchy
- Balance text and white space
- Include one supporting visual
- Maintain consistent formatting
- Ensure readability at presentation distance

Content Guidelines

- Use bullet points, not paragraphs
- Lead with conclusions, not process
- Quantify when possible
- Avoid jargon and acronyms without definition
- Make every word count

Readability Test: Can someone unfamiliar with your project understand your main points in under 2 minutes? If not, simplify and focus.

TASK 8

Peer Review

What Your Partner Will Check

Peer review provides critical feedback before final submission. Your reviewer acts as a fresh set of eyes, catching issues you might have missed and suggesting improvements to strengthen your presentation.

Clarity of Explanation

Can they understand your findings without additional explanation? Is your logic clear and easy to follow?

Logic of Risks

Do your risk assessments make sense? Are likelihood and impact ratings justified with clear reasoning?

Safety-First Thinking

Does your analysis prioritize operational safety and system integrity over technical showmanship?

Slide Readability

Is your one-slide presentation visually clear, well-organized, and understandable at a glance?

After receiving peer feedback, you will refine your work. This iterative process mirrors real professional practice, where drafts are reviewed and improved before stakeholder presentation.

> **Feedback Mindset:** View peer review as an opportunity, not criticism. The goal is to make your work stronger, not to prove you were already perfect.

What This Project Is NOT

✗ This Project is NOT about:

Hacking Systems

No unauthorized access, exploitation, or offensive security activities

Breaking Controls

No attempts to bypass security measures or demonstrate system vulnerabilities

Writing Exploits

No malware development, exploit code, or attack tool creation

Proving Technical Superiority

Not a competition to show advanced technical skills or outperform peers

✓ It is about:

Safe Reasoning

Analyzing systems with responsibility, understanding consequences of findings

Clear Documentation

Creating professional-grade records that support decision-making

Responsible Communication

Presenting security information appropriately to relevant stakeholders

Defensive Understanding

Building knowledge to protect systems, not attack them

Professional Ethics: Real OT security professionals balance technical capability with ethical responsibility. This project builds that mindset from day one.

SUBMISSION READINESS

Submission Readiness Check

Before Submitting, Confirm:

1

Asset Inventory Completed

All assets documented with name, type, role, criticality, and ownership. Inventory is thorough and well-organized.

2

Risk Register (Minimum 5 Risks)

At least five risks identified with justified likelihood, impact, and practical mitigation strategies for each.

3

Monitoring Summary Written

Analysis of dataset completed, addressing what happened, when, where, and why it matters. Findings are evidence-based.

4

One-Slide Presentation Ready

Final presentation slide designed, incorporating system context, approach, findings, and recommendations. Readable in under 2 minutes.

5

File Names and Folders Clean

Project structure follows specified organization. Files are properly named, folders are logically arranged, and everything is easy to navigate.

Critical Reminder: Incomplete structure equals incomplete project. A partially organized submission suggests incomplete thinking. Professional presentation matters as much as technical accuracy.

Thank You

Next: MODULE 12 — Assessment Day

MCQ Examination

Multiple-choice assessment covering concepts from all modules

Mini-Project Presentation

Present your one slide in 2 minutes, demonstrating clear communication

Clarification Questions

Short discussion to verify understanding and reasoning behind your choices

Be Clear

Communicate your findings and reasoning with precision and professionalism

Be Honest

Present what you know, acknowledge what you don't, and avoid overreaching conclusions

Be Confident

Trust your preparation, demonstrate your learning, and present with conviction

You've learned the concepts. You've practiced the skills. Now you're ready to demonstrate your understanding. Approach assessment day with the same professionalism and safety-first mindset you've developed throughout this course.